SWEN30006
Software Modelling and Design

Use Cases

Textbook: Larman Chapter 6 & 30

"The indispensable first step to getting the things you want out of life: decide what you want."

— Ben Stein

Learning Objectives

On completion of this topic, you should be able to:

- Recognise and interpret use cases in their different forms and use case diagrams
- Relate use cases with *include* and *extend* associations, in both text and diagram formats
- Apply tests to identify suitable use cases

Use Cases

- Use cases are widely used to discover and record requirements
- Use cases **are *text stories*** of some actor using a system to meet goals
- Use cases ***emphasise*** the user goals and perspective: "Who is using the system, what are their typical scenarios of use, and what are their goals?"

Process sale use case for POS: Customer arrives at POS checkout with goods and/or services to purchase. Cashier starts a new sale. Cashier enters item identifier. System records sale line item and presents item description, price, and running total. Price calculated from a set of price rules. *Cashier repeats item entry as required and then indicates done.* System presents total with taxes calculated by a tax calculator external system. Cashier tells Customer the total and asks for payment. Customer pays and System handles payment. System logs completed sale and sends sale and payment information to the external Accounting system (for accounting and commissions) and Inventory system (to update inventory). System presents receipt. Customer leaves with receipt and goods (if any).

See a complete use case of process sale in Chapter 6.8 in Larman's textbook

Definitions

- ***SuD:*** *S*ystem-under-Discussion
- ***Actor***: something with behaviour, such as a person, computer system or organisation (e.g. a cashier).
- ***Scenario*** *(or* ***use case instance****)*: a specific sequence of actions and interactions between actors and the SuD.
- ***Use case***: a collection of related success and failure scenarios that describe an actor using the SuD to support a goal.

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Cashier starts a new sale. Cashier enters item identifier. System records sale line item and presents item description, price, and running total. Price calculated from a set of price rules.
Cashier repeats steps 3-4 until indicates done.
System presents total with taxes calculated by a tax calculator external system.
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Scenario
(use case instance)

Actor

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Three kinds of actors

- ***Primary actor*** has user goals fulfilled through using services of the SuD (e.g. the cashier).
  - User goals drive the use case
- ***Supporting actor*** provides a service (e.g. information) to the SuD to clarify external interfaces and protocols (e.g. an automated payment authorization service)
  - Usually be a computer system but could be an organization or person.
- ***Offstage actor*** has an interest in the behaviour of the use case, but is not primary or supporting (e.g. a government tax agency).
  - Ensure all interests identified/satisfied

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Process sale use case for POS: Customer arrives at POS checkout with goods and/or services to

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Example: Handle Returns (casual)

A main success scenario for Handle Returns use case for POS

A customer arrives at a checkout with items to return. The cashier uses the POS system to record each returned item ...

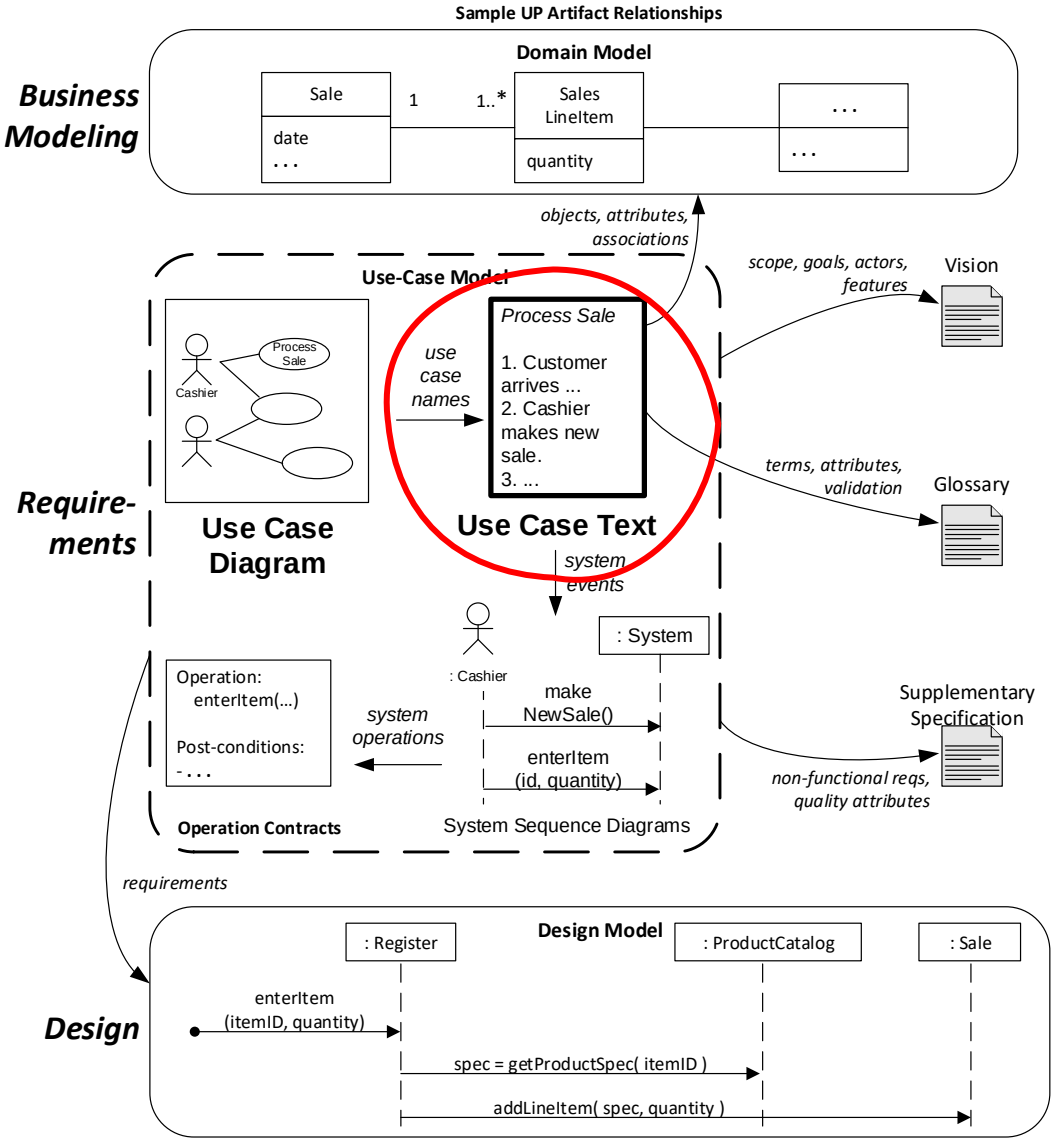
Alternative scenarios

If the customer paid by credit, and the reimbursement transaction to their credit account is rejected, inform the customer and pay them with cash.

If the system detects failure to communicate with the external accounting system, ...

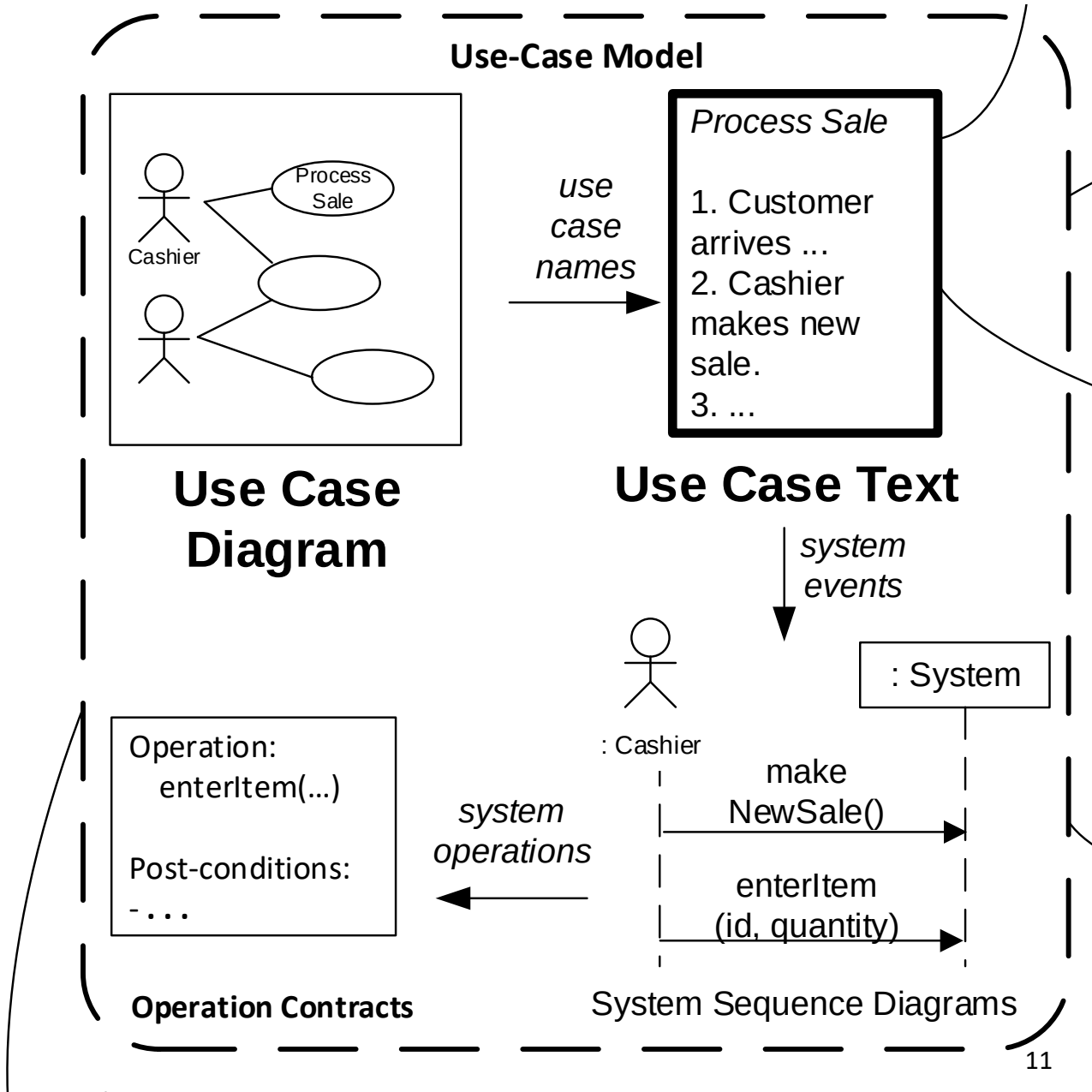
Use Case Influence in Design and Development

- The use cases influence many aspects of a software development project
  - Design
  - Implementation
  - Project Management
- Use cases are key source of information of OO analysis and testing
- Use cases should be strongly driven by the goals of the project



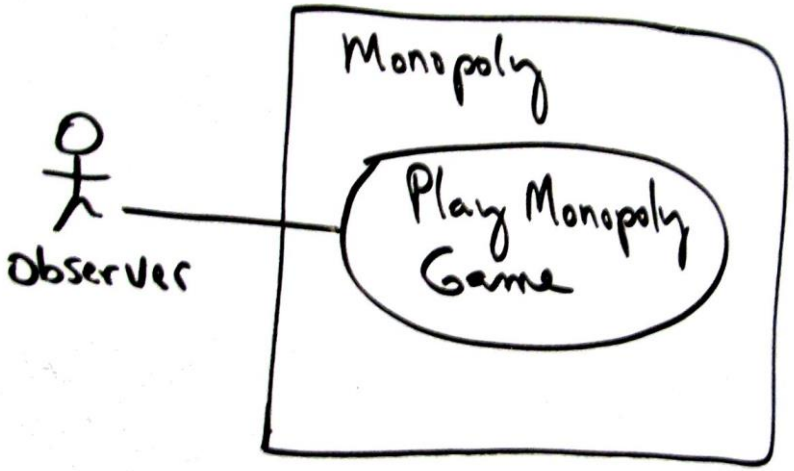
Use-Case Model

- Use-Case Model: A model of the system's functionality and environment
  - *Primarily* includes the set of all written use cases
  - *Optionally* includes a UML use case diagram

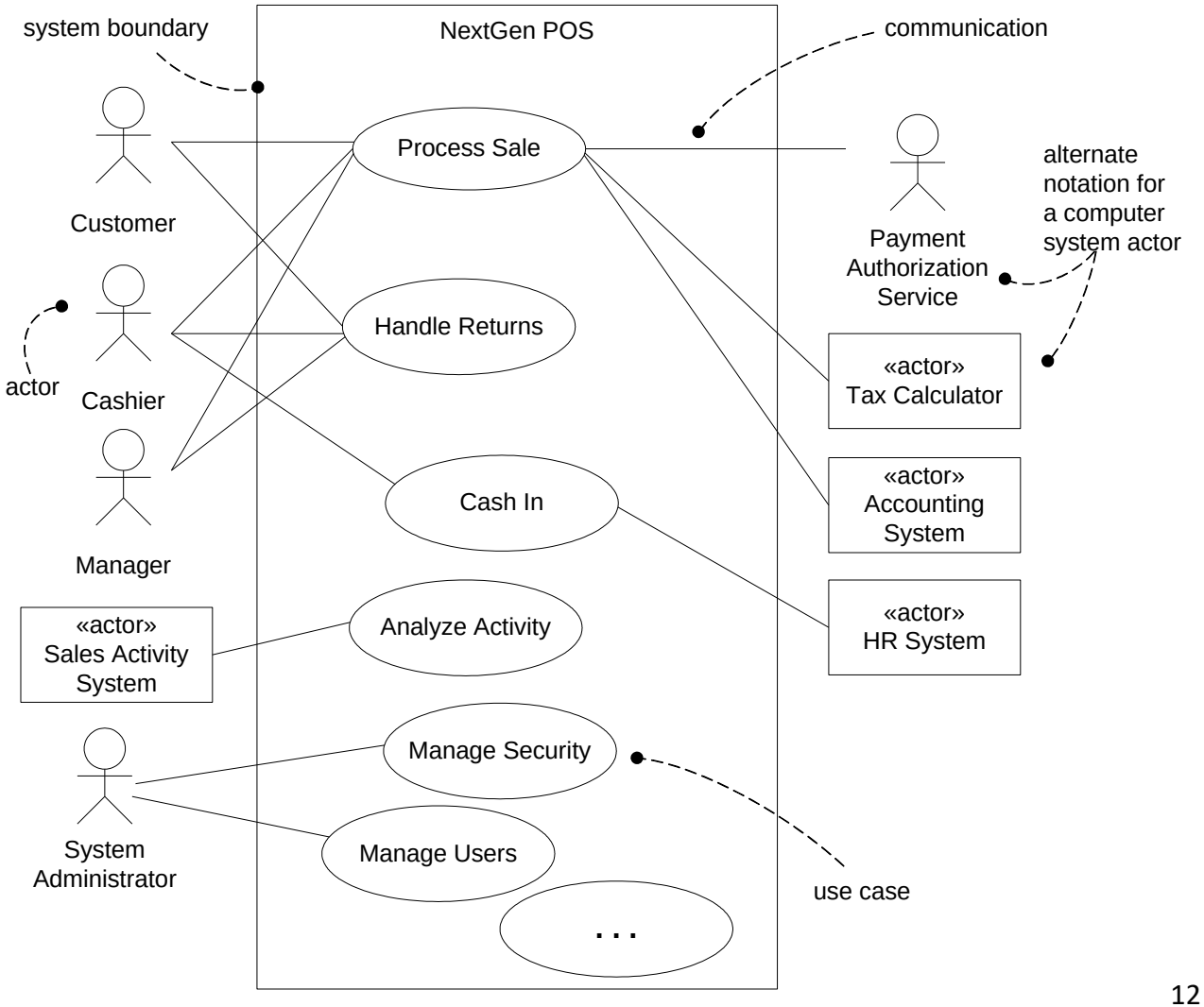


UML use-case diagram

- A UML use case diagram gives a context diagram of a system and its environment
  - Showing the names of use cases, actors and their relationships



A UML use case diagram for Monopoly Simulation System



A UML use case diagram for POS

See Workshop 1 pre-learning module for UML notations and examples

Finding Useful Use Cases

Finding Useful Use Cases

*"What is a useful level to **express** use cases for application requirements analysis?"*

- Boss Test
  - If your boss asks: "What have you been doing all day?" and you reply: *"Logging in!"*, will your boss be happy?
  - Is "Log In" use case useful?
- Elementary Business Process (EBP) Test
  - A task performed by one person in one place at one time, in response to a business event, which *adds measurable business value* and leaves the data in a consistent state
  - Does a use case like "*delete a line item"* reflect EBPs? What about *"Handle Returns"*?
- Size Test
  - Is a task very seldom a single action/step; typically many steps; fully dressed often require 3–10 pages of text

Example: Applying Tests

- **Negotiate a Supplier Contract**
  - Much broader and longer than an EBP. Could be modelled as a *business* use case, rather than a system use case.
- **Handle Returns**
  - OK with the boss. Seems like an EBP. Size is good.
- **Log In**
  - Boss will not be happy if this is all you do all day!
- **Move Piece on Game Board**
  - Single step—fails the size test.

Summary & Remarks

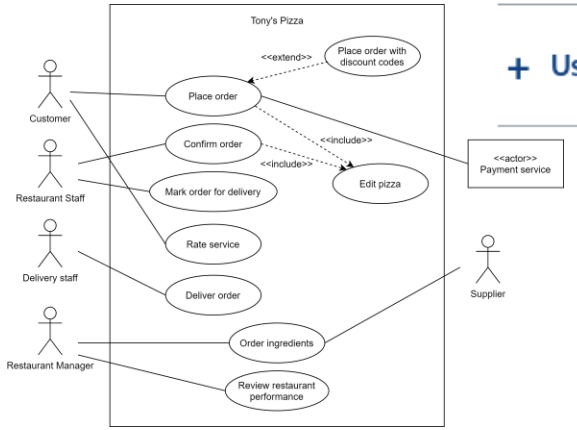
- ***Use Cases* are text descriptions** of an *actor* using the system to achieve a goal.
- ***Use Case Diagrams*** provide a context of use cases and show some relationships between use cases and actors
- Use Cases should cover both success and failure scenarios.
- Use cases can be at different levels of detail.
  - Use the tests to identify the useful use cases
- Need more examples of use cases?
  - Checkout Tony's Pizza Case study in LMS!

+ Use Case 1: Order ingredients (Brief style)

+ Use Case 2: Deliver order (Casual style)

+ Use Case 3: Place order (Fully dressed style)

+ Use Case 4: Edit Pizza (Fully dressed style)



Lecture Identification

Lecturer: Philip Dart

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These slides include materials from:

Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development, Third Edition, by Craig Larman, Pearson Education Inc., 2005.